

An inverse function

An inverse function reverses a one-to-one function: outputs of the original become inputs of the inverse.

Why must the original function be one-to-one?

Inverse notation

The superscript -1 in inverse-function notation names the inverse.

$$f^{-1}(x)$$

How is this different from the reciprocal $1/f(x)$?

Reverse an input-output pair

$$f(3) = 11$$

What value must $f^{-1}(11)$ have if f has an inverse?

A finite inverse

$$f = \{(1, 4), (2, 7), (3, 9)\}$$

What ordered pairs represent its inverse?

When the inverse relation fails

$$f = \{(1, 5), (2, 5), (3, 8)\}$$

Why is the inverse relation not a function?

Two-way verification

Inverse functions undo each other on their respective domains.

$$f(g(x)) = x, \quad g(f(x)) = x$$

Why should both compositions and their domains be checked?

Verify a linear inverse

Use $f(x) = 4x - 7$ and $g(x) = (x + 7)/4$.

Are f and g inverse functions?

A failed inverse claim

Use $f(x) = 2x + 3$ and $g(x) = (x - 2)/3$.

Are f and g inverse functions?

Find a linear inverse

$$f(x) = 5x + 2$$

What formula represents f^{-1} ?

Find an inverse with fractions

$$f(x) = \frac{x - 4}{3}$$

What formula represents f^{-1} ?

A rational inverse

$$f(x) = \frac{2}{x-1}$$

What are the inverse's formula and domain?

Swap domain and range

A one-to-one function has domain $[2, 6]$ and range $[-3, 5]$.

What are the domain and range of its inverse?

Reflect a graph

An inverse graph swaps the coordinates of every point.

$$(a, b) \longleftrightarrow (b, a)$$

Why is the line $y = x$ the line of reflection?

Plot inverse points

A one-to-one function's graph includes $(-2, 1)$, $(0, 3)$, and $(4, 7)$.

Which points must appear on the graph of its inverse?

A quadratic is not one-to-one

$$f(x) = x^2$$

Why can this function not have an inverse on all real inputs?

Restrict a quadratic domain

Use $x \geq 0$.

$$f(x) = x^2$$

What inverse formula and domain result?

Choose the other branch

Use $x \leq 0$.

$$f(x) = x^2$$

How does the inverse differ from the nonnegative-domain case?

A shifted quadratic inverse

Use $x \geq 3$.

$$f(x) = (x - 3)^2 + 2$$

What is the inverse, including its domain?

An inverse square-root function

$$f(x) = \sqrt{x - 4} + 1$$

What inverse formula and domain reverse this function?

An inverse cube

$$f(x) = (x + 2)^3 - 5$$

What is its inverse function?

Reverse a model

The total printing cost in dollars is $C(n) = 0.20n + 8$, for whole numbers $n \geq 0$.

What does the inverse model's input represent?

Use the inverse model

The total printing cost in dollars is $C(n) = 0.20n + 8$, for whole numbers $n \geq 0$.

How many pages correspond to a cost of \$18?

Domain-sensitive verification

Use $f(x) = x^2$ on $x \geq 0$ and $g(x) = \sqrt{x}$ on $x \geq 0$.

$$g(f(x)) = \sqrt{x^2}$$

Why does this composition equal x on the stated domain?

Reverse an exponential

Use a graphing utility.

$$f(x) = 2^x$$

What domain and range would its inverse have?